Nachrichtenbl. Deut. Pflanzenschutzd., **59** (7), pp. 155–161, 2007, ISSN 0027-7479.

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Definition of soil-climate zones for the Federal Republic of Germany Germany

Definition of soil-climate areas for Germany

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Summary

The regional classification presented in the following article "Soil-climate regions of the Federal Republic of Germany" presents a consensus between the agricultural institutions/offices or the chambers of agriculture of the federal states and the Federal Biological Research Centre for Agriculture and Forestry The participating authorities will use the soil-climate zones (SCLs) defined therein as building blocks for creating further, more specific, "coarser" regional classifications that meet particular objectives. Due to the clear delimitation of the SCLs based on municipal boundaries,

This also enables a clear assignment of each agricultural operation, thus simplifying administrative management and This facilitates, or even makes possible, the handling of this regional classification. This is particularly true for applications related to geographic information systems (e.g., presentation of BKR-related key figures). (as cards). "Last but not least," a practical Work-complicating parallel developments regarding agricultural regional divisions should be avoided (at least as far as the involved institutions).

Through the scientific approach to defining the BKR and the intensive coordination with those responsible in the individual federal states, the authors hope for a high level of acceptance for the presented regional division.

Keywords: Regional division, soil-climate regions, geographic information systems

Abstract

The presented area segmentation "soil-climate-regions of the Federal Republic of Germany" is a consent of the authorities for agriculture and the chambers of agriculture of the "Bundeslän-der" and the Federal Biological Research Center for Agriculture and Forestry. The mentioned authorities will use the defined soil-climate-regions (scr) as elements of further area segmentation, in particular for purposes using a coarser segmentation. The scr are clearly designated on the basis of community borders. For this reason it is possible to clearly assign agricultural enterprises and hence to simplify administration and handling of area segmentation or to make it possible at all. This is especially true of the use of geographic information systems (for instance presentation of scr-related indices as maps). And it was possible to avoid parallel lel developments in agricultural area segmentation (at least among the participating authorities). Such a development would have made practical work more difficult.

The authors hope for high acceptance of the presented area segmentation because of their scientific approach in designating scr and intensive agreement between the responsible parties.

Key words: Geographical classification, soil-climate areas, geographical information systems

1 Introduction

For the Federal Republic of Germany, there are a large number of Territorial divisions. The best-known and probably most frequently used is the division into federal states, administrative districts, counties, and municipalities. However, there are also territorial divisions without any reference to such administrative aspects (e.g., geological maps, natural region classifications). and similar things).

As a general rule: Every definition of sub-areas depends on the specific objective and the available data regarding... the influencing factors to be considered and the methodology used for this purpose. As a rule, therefore, a new one always emerges. standalone map.

This also applies to the agricultural sector, of course. In recent years, the working group "Coordination in Experimentation" at the Association of Chambers of Agriculture, in which all state agencies responsible for variety testing are represented, has developed a classification. The Federal Republic of Germany developed soil-climate regions with the aim of optimizing the implementation and evaluation of variety trials and variety advisory services. In parallel, the Federal Biological Research Centre for Agriculture and Forestry defined soil-climate regions for its statistical surveys on the use of plant protection products in arable farming. (NEPTUN project). The latter were submitted to the official plant protection services of the federal states for comment. Since in the majority of the federal states plant protection services and variety testing are under the joint umbrella of the project, the following measures were taken: Roof of a state agricultural institute or a rural- The fact that the chambers of commerce are organized resulted from the existence of two different regional divisions (in addition to this). (with very similar names) on the one hand, certain irritations and on the other hand, the desire to coordinate the corresponding cards with each other.

In a working session convened for this reason, decided:

- Coordination between the various institutions is supported.
- The aim is to delineate areas with relatively homogeneous site conditions for agricultural production.
  It is particularly important to consider the different influences of soil quality and climate. Therefore, these sub-areas should also be referred to as soil-climate regions (SCR).
- The number of soil climate spaces to be defined should between 50 and a maximum of 80.
- These BKR then represent the units from which larger, sub-areas adapted to the respective purpose (regions) are composed (in variety trials, for example, fruit-specific cultivation or advisory areas; in plant protection NEPTUN survey regions arable farming).

2 Methodology, Results and Discussion

The separation of methodology, results, and discussion, which is common in scientific publications, is deliberately omitted here because it obscures the individual work steps and the resulting partial results can be worked out or presented much more clearly, and the entire development process is easier for the reader to understand.

2.1 Area reference

With regard to the later uses of the area classification to be developed, it was decided to define the soil-climate zones on the basis of municipal boundaries. Only This allows for the practical use of this regional division for the This ensures the fulfillment of various tasks. Additionally, this approach simplifies administrative implementation. (Assignment of agricultural holdings to the BKR, collection and management of BKR-related data, etc.).

Furthermore, this determination also corresponded to the circumstances regarding the available digitized geometric data (GEM 2000; a product of the company ESRI1).

Software tools (ArcGIS, ArcInfo,

ArcMap2) from the company ESRI.

2.2 Influencing factors

The two most significant influencing factors of location on the Agricultural production depends on soil and weather; in particular temperature and precipitation.

To quantify the influencing factor of soil, the Map "Indicative soil types of Germany" (Fig. 1), which was created by the Federal Institute for Geosciences and Natural Resources, used. Each of these guide soils was evaluated using a table. (SCHULZKE, unpublished) assigned a land value rating. The associated, sometimes considerable, loss of information Unfortunately, this was unavoidable. Due to data overlay.

Using the municipal boundaries, a weighted average soil quality could be calculated for each municipality (Fig. 2).

The quantification of the influencing factor weather was carried out through climatological values. This resulted in effects. from random, year-specific weather peculiarities avoided in the definition of the BKR. Data from 401 meteorological stations (hereinafter referred to as climate stations) (named) are available. Each municipality was assigned a climate station representative of that municipality. (Fig. 3).

Since the requirements for temperature and water supply (precipitation) vary both quantitatively and in terms of timing depending on the fruit species, a compromise had to be found for their inclus These two meteorological factors will be considered. It was decided to use the mean monthly temperatures. and the average monthly rainfall totals to use the period from March to August (Figs. 4 and 5).

2.3 Cluster formation

In the next step , the municipalities were identified using a clustering procedure (SAS3 ; proc cluster; method=ward). with similar properties regarding soil quality, temperature and Precipitation is grouped into larger areas (clusters). A prerequisite for a comprehensible grouping was the Restriction to the three largely uncorrelated variables. To compensate for the different scale levels and as a prerequisite for weighting the variables equally, a Standardization. The evaluation of various quality criteria. (Cubic Clustering Criterion [CCC]; Root Mean Square Standard Deviation [RMSSTD]) yielded, taking into account the specification, "so Forming clusters "as few as possible, but as many as necessary", that a summary into 70 clusters best met the framework conditions. However, as expected, these 70 clusters did not form contiguous areas, but rather... Each patch consisted of a large number of sub-areas (2131 patches). To improve clarity and without To avoid impairing the objective for the territorial division, all patches smaller than were subsequently removed. 20,000 hectares were combined with the neighboring area, which with regard to the The indicator "average soil quality" was most similar. Therefore

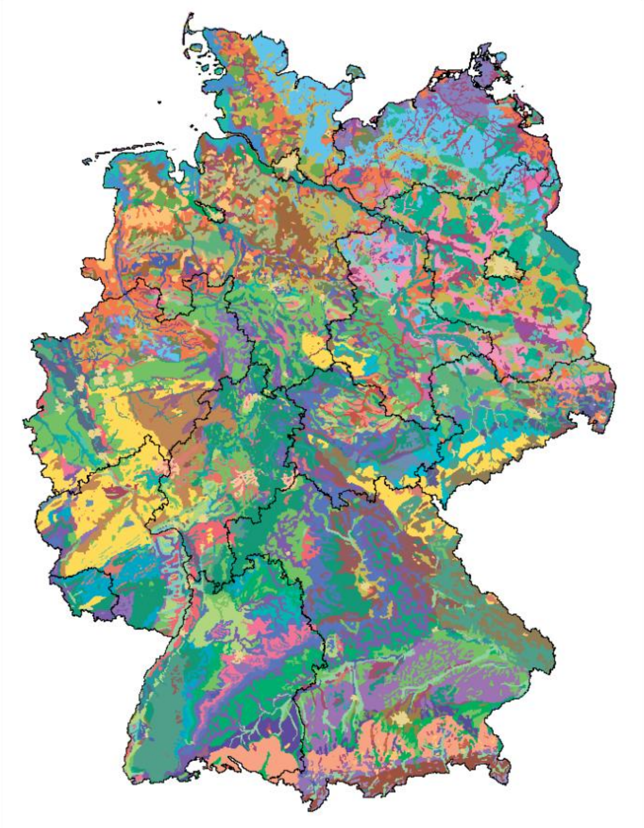

Fig. 1. Typical soil types of Germany.

1) ESRI Geoinformatics GmbH, Kranzberg

2) ArcGIS, ArcInfo, ArcMap Copyright © 1995-2005 ESRI

3) SAS (version 9.1), SAS Institute Inc., Cary, NC, USA

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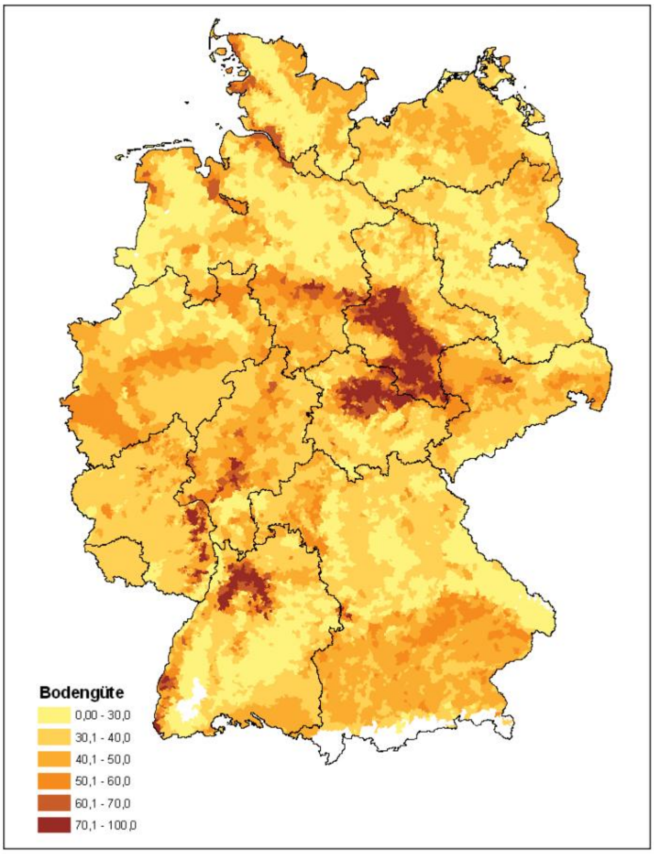

Fig. 2. Average soil quality (based on: indicator soil types).

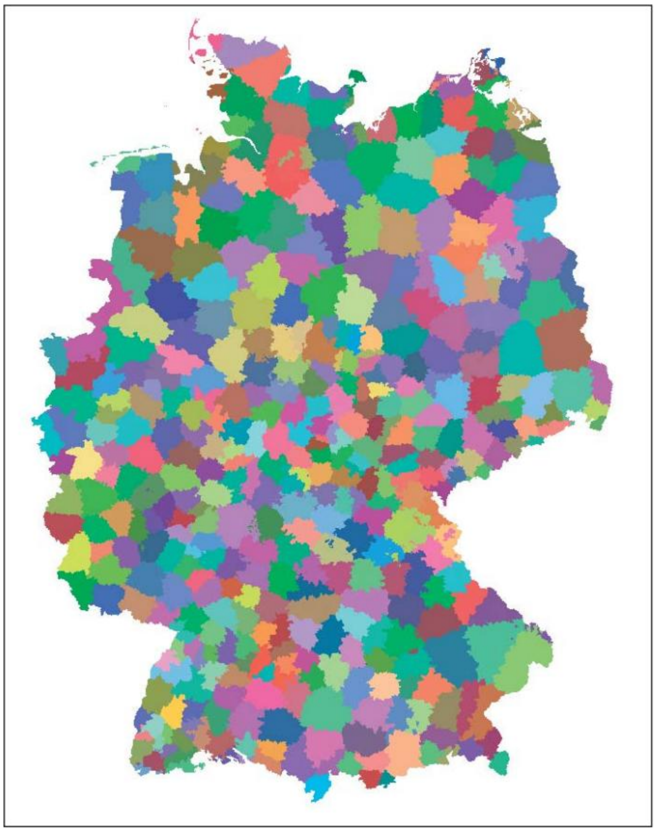

Fig. 3. Regional breakdown for Germany (areas assigned to climate stations on a municipal basis).

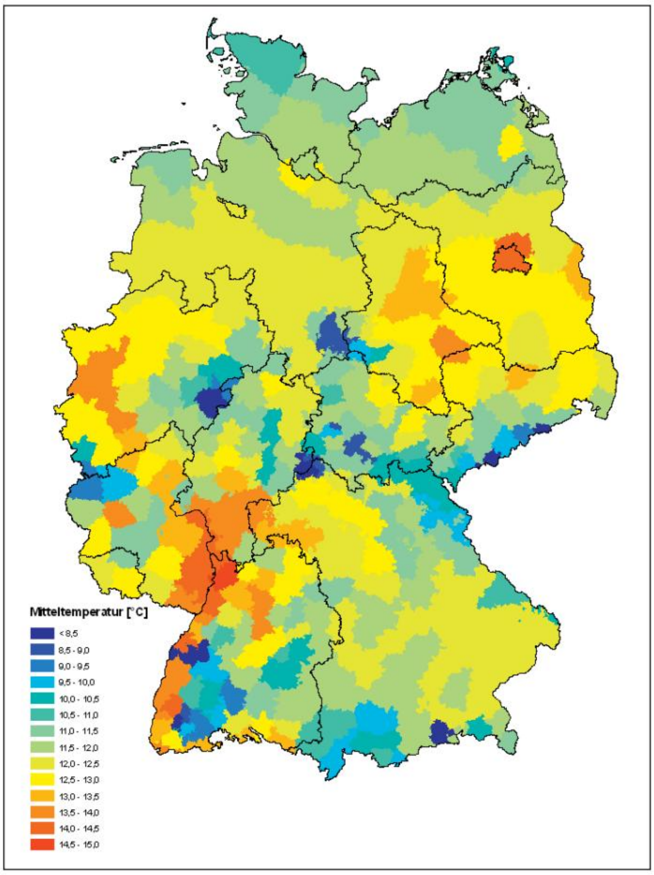

Fig. 4. Average temperature (March to August).

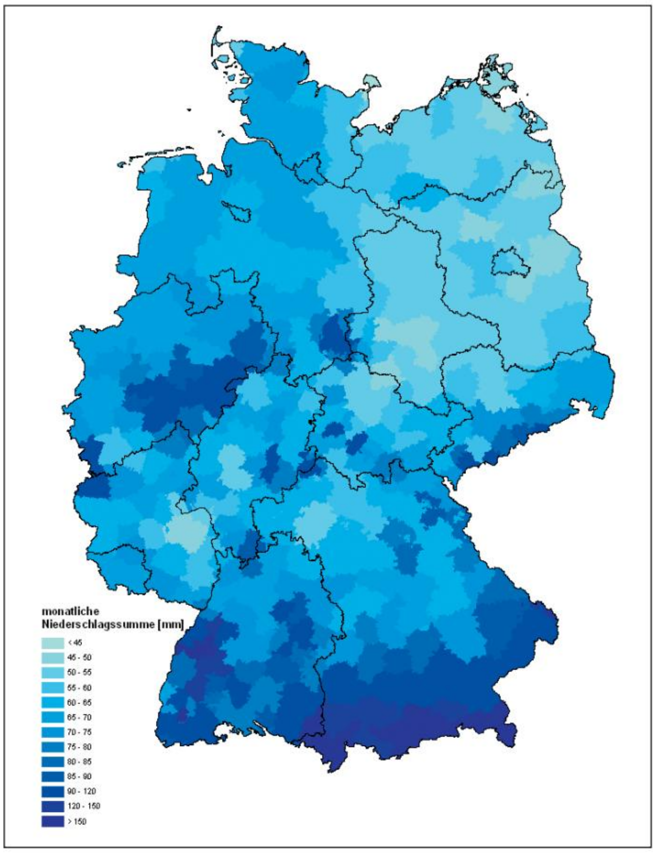

Fig. 5. Average monthly rainfall total (period March to August).

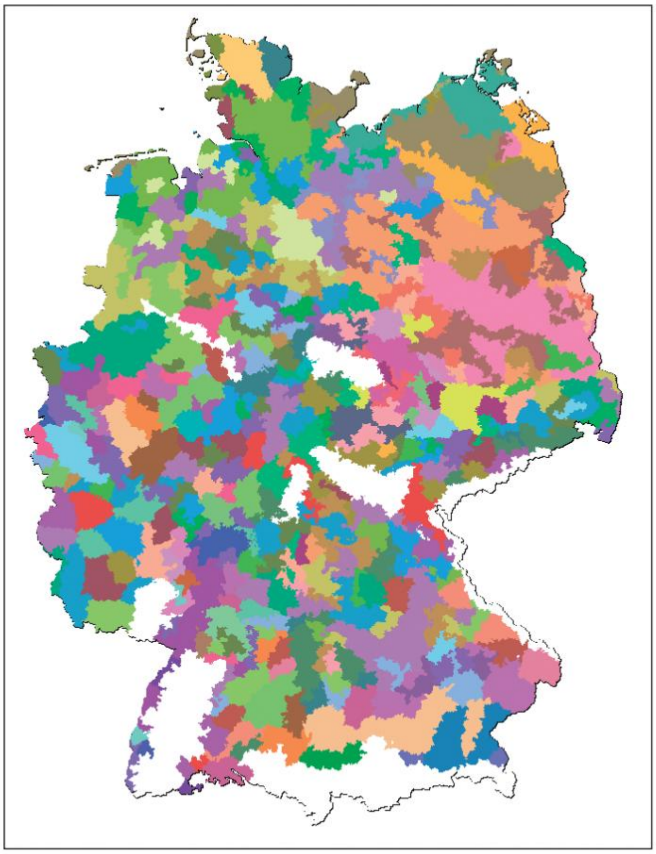

Fig. 6. Visualized result of the cluster analysis.

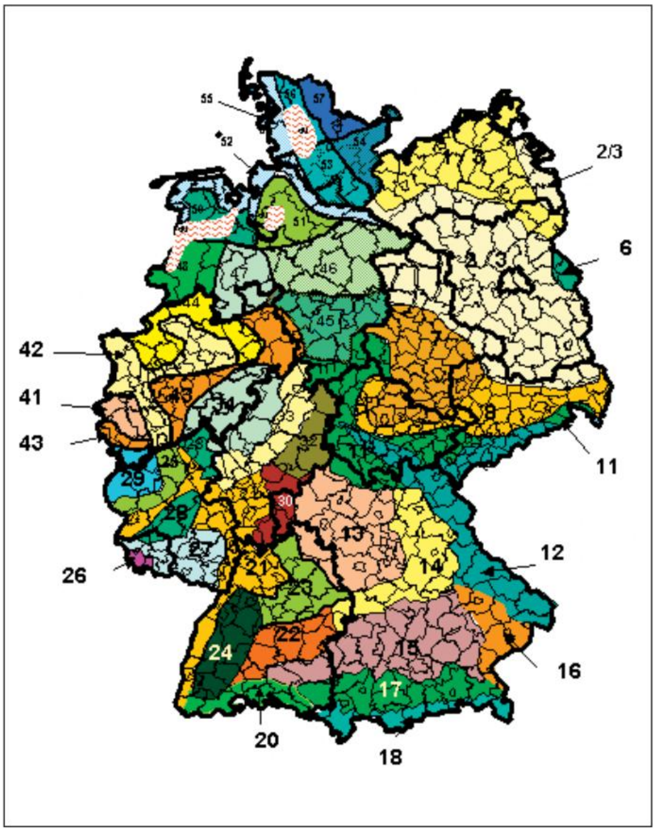

Fig. 7. Soil-climate regions – Germany; Variety trial version being.

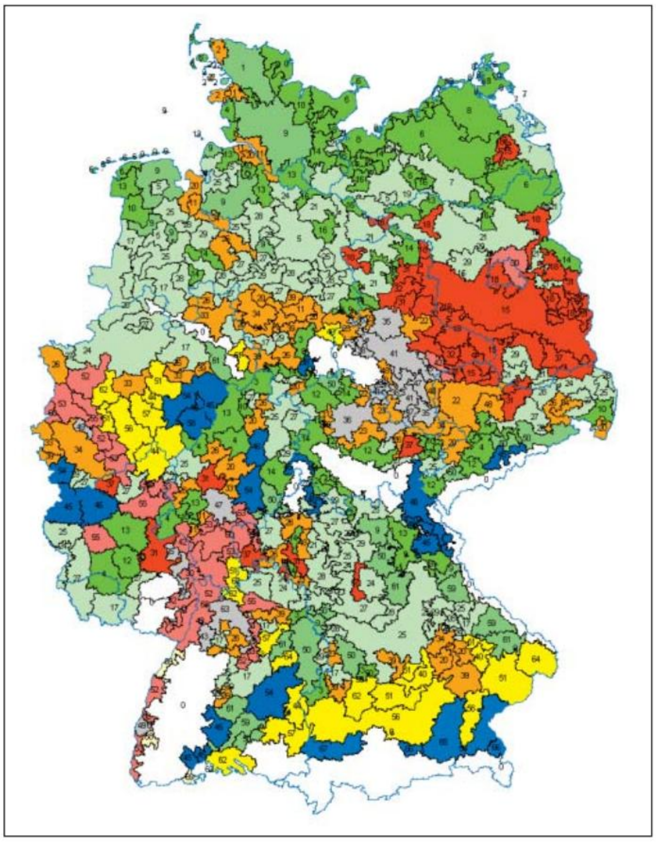

Fig. 8. Visualization of cluster groups.

The number of patches was reduced to 439. The result of clustering and elimination of "SMALL" areas Figure 6 shows this.

2.4 "Preliminary" definition of soil-climate areas

However, the regional division shown in Figure 6 still resembles a "patchwork quilt" and cannot be further refined in this way.
The form could not be used in practice. Therefore, it was necessary to further combine the defined sub-areas (patches).
However, only mathematically oriented methods were available for this purpose.
Driving is only possible to a limited extent or not at all suitable. At this point it was It is necessary to incorporate "expert knowledge" and "on-site knowledge" into the process. The figures in Figure 7 illustrate this.
The map shown represents an implementation of the aforementioned subjective factors. It was developed in the working group "Coordination in "Experimental science" at the Association of Chambers of Agriculture er-

This work is part of a resolution passed by the Conference of Agriculture Ministers on October 7, 2004, concerning the "Reorganization of Variety Trials." Unfortunately, the resources required for the development of the information shown on this map were insufficient.
The depicted regional division does not use uniform parameters. agreed upon for the definition of soil-climate zones. Likewise.
There are no specifications regarding the desired size of sub-areas and the maximum tolerable heterogeneity within the Sub-regions. On the other hand, regional characteristics are adequately reflected.

The second template (Fig. 8) was created by utilizing the The fact that the chosen method for cluster formation at further reduction of the number of clusters step by step grouped together "similar" clusters. Taking into account the Based on the quality criteria mentioned in section 2.3, it was decided to classify the 70 clusters into 11 cluster groups and

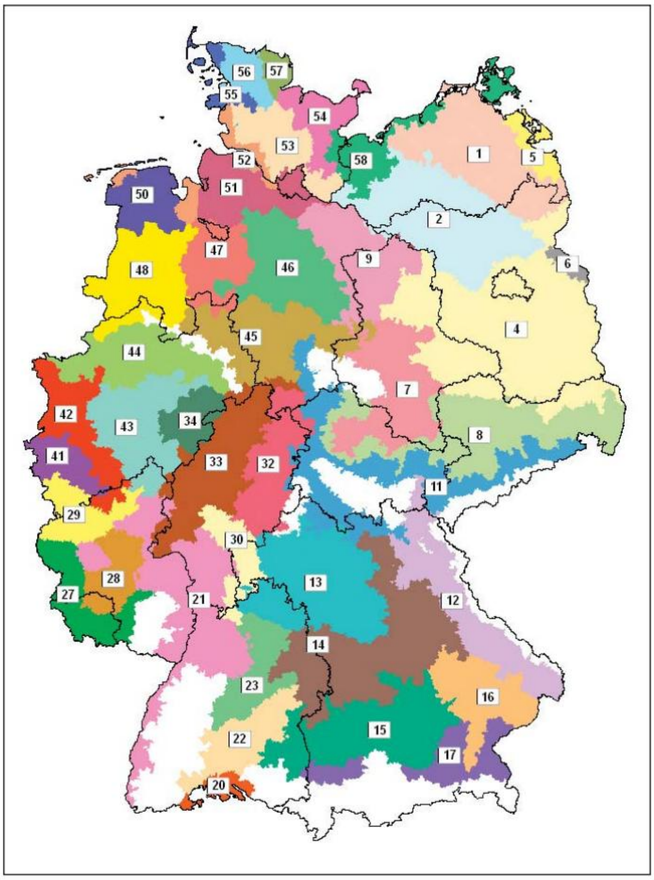

Fig. 9. Soil-climate rooms; design June 2006.

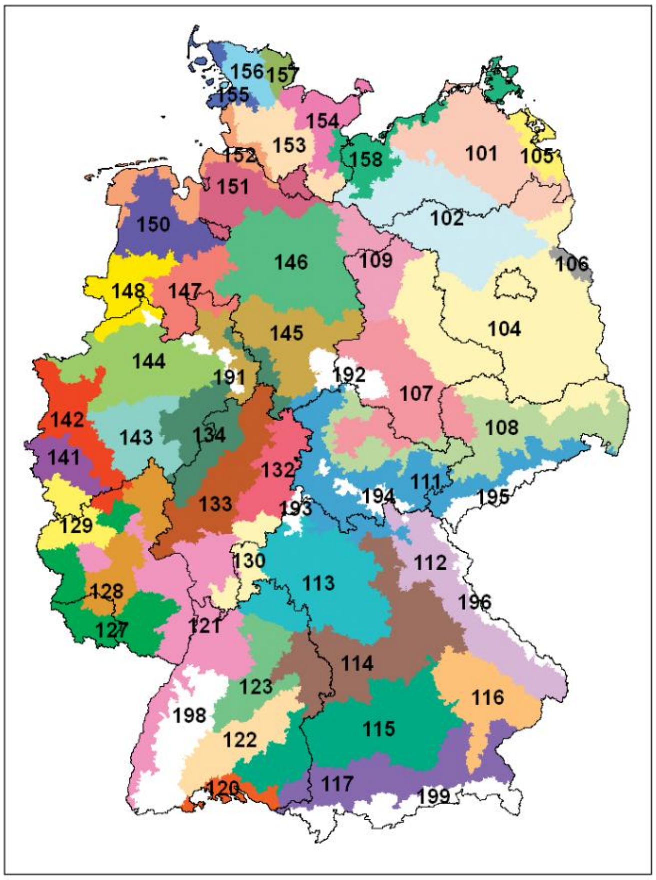

Fig. 10. Soil-climate rooms 2007.

all patches belonging to a cluster group are applied uniformly. "to colorize". In addition, the color scheme was designed in such a way that that "similar" cluster groups with the same base color in different shades are displayed (e.g. three Green tones). This approach led to a far greater Comprehensible visualization of the site-specific conditions in Germany. Similarities to the predominantly regionally based knowledge of the conditions. The regional divisions (Figure 7) become apparent.

Based on the two maps mentioned above, the boundaries of the newly defined soil-climate regions (Fig. 9) were finally established. Figure 9 thus represents a regional division based on measurable, nationwide data.

based on uniform influencing factors and derived using objective mathematical methods; supplemented with regional fine-tuning based on expert knowledge.

2.5 "final" definition of soil-climate spaces

In the final step, the preliminary work completed up to this point was... See the definition of soil-climate zones as a map. (see Fig. 9) as well as a list with the corresponding assignment The individual municipalities forwarded the information to the BKR (Federal Commission for Church Records) and the responsible authorities of the federal states. The relevant experts Those "on site" were asked to compare this regional division with their own experiences and, if necessary, to formulate suggestions for changes that they considered sensible.

This vote, however, led to numerous, sometimes significant, changes to the intended regional structure. The reasons for this were manifold.

Firstly, the authors examined the agricultural use- Interpretation of large regions in the low mountain ranges (in draft: "Excluded" forest areas) are underestimated. Many parts of these Forest areas not originally considered were assigned to adjacent soil climate rooms. In some In some cases, this assignment also represents a compromise between those responsible for arable farming and those responsible for grassland.

Further corrections resulted from the fact that the original draft did not include local geographical features (local elevations, river courses and similar features) not or not sufficient consideration could be given to this. In addition, The classification of municipalities located in the "border regions" of the respective soil-climate areas was frequently changed. In many cases, this also became apparent (especially in the case of larger municipalities), that within a municipal area quite heterogeneous conditions for agriculture Production is encountered, and therefore, in principle, the assignment to different BKR (Business Process Control Units) is technically justifiable. In these cases, the opinion of the local authorities naturally plays a role. Agricultural consultants or research associates have a special role Role.

Suggested changes stemming from "long-standing" habits or administrative circumstances also presented some challenges. Critical questions

However, they helped to limit such changes to a minimum (according to the authors).

Figure 10 shows the result of the agreement on the regional division with the federal states. This division Germany's soil-climate spaces are being studied by all involved Institutions accepted and used for their tasks in the coming years. Table 1 lists the terms for the Individual BKR listed.

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Table 1. BKR designations

BKR number	BKR designation
101	medium diluvial soils MV and Uckermark
102	sandy diluvial soils of northeastern Germany Inland lowlands
104	dry-warm diluvial soils of East Germany lowlands
105	Western Pomeranian sandy soils in the Uecker-Randow region Area
106	Oderbruch
107	Loess soils in the arable plain (East)
108	Loess soils in the transitional areas (East)
109	Diluvial soils of the Altmark and overlap with northern Lower Saxony
111	Weathered soils in the transitional areas (East)
112	Weathered soils in the higher elevations (eastern Bavaria)
113	Northwest Bavaria-Franconia
114	Swabian Alb and East Bavarian Hill Country
115	Tertiary hill country Danube-South
116	Gäu, Danube and Inn Valley
117	Moraine hill country and Alpine foothills
120	Upper Rhine-Lake Constance
121	Rhine plain and side valleys
122	Swabian Alb, Baar
123	Upper Gäu and transitional areas suitable for grain maize
127	Central regions of Rhineland-Palatinate and Saarland
128	Hunsrück, Westerwald
129	High Eifel/High Altitudes
130	Odenwald, Spessart
132	East Hessian low mountain ranges
133	Central Hessian agricultural areas, Warburg Börde
134	Sauerland, Brilon Heights
141	Jülich Börde, Zülpich Börde/lowlands, moist
142	Upper Middle Rhine, Lower Rhine, southern Münsterland
143	East Westphalia, Lippe, Haarstrang, Bergisches Land/ Transitional areas, moderately humid
144	Münsterland/lowlands, dry
145	Southern Hanover/Clay soils
146	Lüneburg Heath/Sandy soils
147	central Lower Saxony/light loamy soils
148	southwestern Weser-Ems region/sandy soils
150	northwestern Weser-Ems region/sandy soils
151	Elbe-Weser Triangle/sandy soils
152	Lower Saxony coastal and Elbe marshes
153	Geest – South
154	southern Schleswig-Holstein hill country
155	March – North
156	Geest – North
157	northern Schleswig-Holstein hill country
158	NW Mecklenburg and coastal area/better diluvial soils
191	Teutoburg Forest
192	resin
193	Rhön
194	Thuringian Forest
195	Ore Mountains
196	Bavarian Forest
198	Black Forest
199	Alps
additional floor climate rooms	
103	Lowland sites in northeastern Germany (predominantly bogs)
160	Moors Northwest Germany

(These locations are small and/or unconnected within arable farming areas and are therefore not shown on the map.)
(shown.)

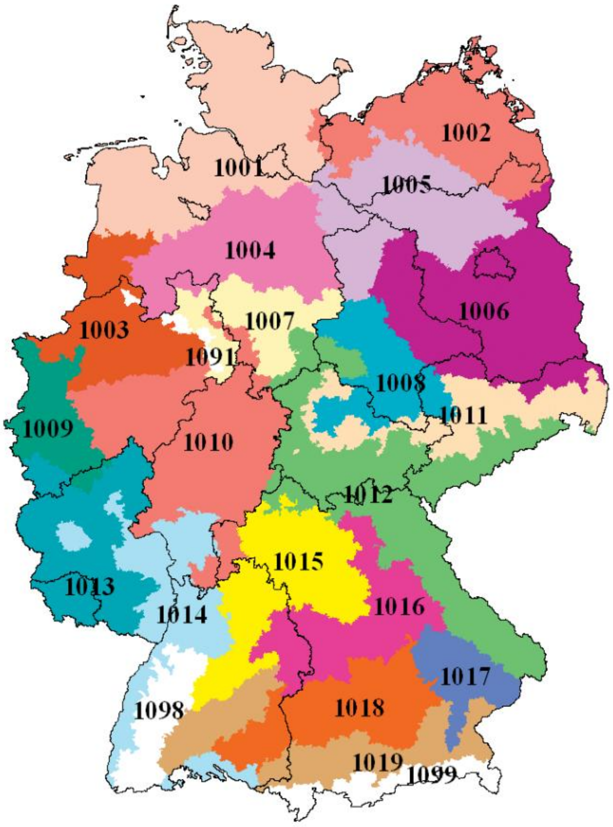

Fig. 11. NEPTUN survey regions: agriculture.

Table 2. ERA designations

ERA number	ERA designation
1001	Schleswig-Holstein/northern Lower Saxony
1002	Mecklenburg-Western Pomerania (excluding the Mecklenburg Lake District)
1003	Westphalian Lowland Basin/western Lower Saxony
1004	Weser-Aller Lowlands/Counties of Diepholz and Hoya
1005	Lüneburg Heath/Altmark/Prignitz/southern Mecklenburg
1006	East and South Brandenburg Heath and Lake District/Fläming
1007	Weserbergland
1008	Central German Black Earth Region
1009	Lower Rhine Bay/Cologne-Aachen Bay
1010	Hessian Central Uplands/Sauerland/ Bergisches Land
1011	Upper Lusatia/Saxon-Thuringian Hill Country
1012	Central German uplands and foothills/ Eastern Bavaria
1013	Saarland/Hunsrück/Eifel
1014	Rhine-Main Lowlands/Wittlich Basin
1015	Northern Gäu Plateau/West Franconia
1016	Keuper-Lias region
1017	Lower Bavarian hill country
1018	Southern German gravel slabs
1019	Bavarian Pre-Alpine Hill Country/Swabian Alb
Forest areas NOT used for NEPTUN surveys	
1091	Teutoburg Forest
1098	Black Forest
1099	Alps

2.6 Aggregation of the NEPTUN survey regions arable farming (ERA)

As conceptually planned, survey regions for the statistical recording of the intensity of pesticide use (NEPTUN project) were subsequently generated based on the soil-climate zones. The survey regions to be used in the future are shown in Figure 11;

the ERA designations in Table 2.

The survey regions represent in the original Areas with comparable natural conditions for agricultural production are desirable. Such regionalization is necessary because the intensity of plant protection measures in an agricultural operation objectively also from regional perspectives Influencing factors such as climate and soil conditions, pest and disease occurrence, and yield expectations are determined. However, the following framework conditions also had to be considered when defining the survey regions:

- Small-scale inhomogeneities could and can occur in the The chosen level of abstraction is not taken into account.
- All survey regions should be contiguous areas be.
- The areas must have a certain minimum size, to determine the necessary sample size (fruit-specific) Data from at least 30 businesses per survey region to ensure the anonymity of the companies!).

Therefore, it was necessary to make significant compromises in defining the ERA.

2.7 Aggregation of Cultivation areas in Variety management

The resolution of the Conference of Agriculture Ministers of October 7, 2004, obliges the responsible state agencies for variety management to provide variety information and recommendations on a "regionalized" or "growing-area-specific" basis. The soil-climate zones form the basis for this independently.

The smallest indivisible unit in terms of its intended use. are the basis for the aggregation of fruit-specific data Cultivation areas. Depending on the distribution and economic importance of the plant species, as well as taking into account the differentiated interactions between variety and environment, A growing area comprises one or more soil-climate regions. The new biometric method of the "Hohenheim-Gülzow series evaluation" is intended to further improve the regional evaluation of variety trials in dynamically overlapping growing areas.

Due to the very different aggregation of cultivation areas between plant species and the additional component of mutual overlapping areas, a

The description of the growing areas is omitted here.

Accepted for publication: March 12, 2007

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